

# Supplementary Material for: ‘Probabilistic Modeling, Learnability and Uncertainty Estimation for Interaction Prediction in Movie Rating Datasets’

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## A Proof of Mathematical Results

In this appendix, we show that low-rank PMFs are learnable in the sense of  $L^1$  norm and further show that this implies an analogue of an excess risk bound in our implicit feedback context: there exists an algorithm which, consuming fewer than  $\tilde{O}((m+n)r/\epsilon^2)$  samples, picks a low rank distribution whose expected recall at  $k$  is guaranteed to be within  $\epsilon$  of the best possible recall at  $k$  achievable.

### A.1 Relating the $L^1$ Loss to the Recall at $k$

Let  $p \in \mathbb{R}^n$  (resp.  $\hat{p} \in \mathbb{R}^n$ ) be a distribution over  $[n]$  (which, as in standard notation, stands for  $\{1, 2, \dots, n\}$ ). We write  $p_{[i]}$  (resp.  $\hat{p}_{[i]}$ ) for the  $i$ th element of  $p$  (resp.  $\hat{p}$ ) when written in decreasing order. We also write  $\sigma$  (resp.  $\hat{\sigma}$ ) for the permutation of  $[n]$  such that  $p_{[i]} = p_{\sigma(i)}$  (resp.  $\hat{p}_{[i]} = \hat{p}_{\hat{\sigma}(i)}$ ).

If we draw a test (multi) set  $\Omega' = \{y_1, \dots, y_{N'}\} \subset [n]$  consisting of  $N'$  i.i.d. samples from  $p$ , the *Recall@k* of a scoring function  $p$  or  $\hat{p}$  is defined as the number of test samples belong to the top  $k$  items as determined by the scoring function  $p$  or  $\hat{p}$ :

$$R_{N'}^k := \frac{1}{N'} \sum_{o=1}^{N'} 1_{y_o \in \sigma^{-1}([k])}, \quad (1)$$

$$\hat{R}_{N'}^k := \frac{1}{N'} \sum_{o=1}^{N'} 1_{y_o \in \hat{\sigma}^{-1}([k])}. \quad (2)$$

This is a random variable. Note that by the i.i.d. assumption its expectation doesn't depend on  $N'$  and is calculated as follows:

$$\mathbb{E}(R_{N'}^k) = \mathbb{E}(R_1^k) = \sum_{i \in \sigma^{-1}([k])} p_i, \quad (3)$$

$$\mathbb{E}(\hat{R}_{N'}^k) = \mathbb{E}(\hat{R}_1^k) = \sum_{i \in \hat{\sigma}^{-1}([k])} p_i. \quad (4)$$

By abuse of notation, we write  $\mathbb{E}(R^k)$  for  $\mathbb{E}(R_1^k)$  and  $\mathbb{E}(\hat{R}^k)$  for  $\mathbb{E}(\hat{R}_1^k)$ .

The quantity  $\mathbb{E}(R^k)$  represents the **best possible expected recall**, and is analogous to the Bayes Error in classic Learning Theory.  $\mathbb{E}(\hat{R}_{N'}^k)$  is the true expected recall of the trained model  $\hat{p}$ , thus, the quantity  $\mathbb{E}(\hat{R}_{N'}^k) - \mathbb{E}(R^k)$  is analogous to the excess risk in learning theory.

We also define the (empirical) estimated recall at  $k$  as follows:

$$\hat{\mathbb{E}}(\hat{R}_1^k) = \sum_{i \in \hat{\sigma}^{-1}([k])} \hat{p}_i = \sum_{i \leq k} \hat{p}_{[i]}. \quad (5)$$

We will now prove the following:

PROPOSITION A.1. *If  $\|p - \hat{p}\|_1 \leq \epsilon$  for some  $\epsilon > 0$ , then we have:*

$$\mathbb{E}(R^k) - \epsilon \leq \hat{\mathbb{E}}(\hat{R}^k) \leq \mathbb{E}(R^k) + \epsilon. \quad (6)$$

*In particular, since we certainly have  $\mathbb{E}(\hat{R}^k) \leq \hat{\mathbb{E}}(\hat{R}^k) + \epsilon$ , we also have the following bound on the excess risk:*

$$\mathbb{E}(\hat{R}^k) - \mathbb{E}(R^k) \leq 2\epsilon. \quad (7)$$

PROOF. We can rewrite the quantity  $\mathbb{E}(R^k) = \sum_{i \leq k} p_{[i]}$  as  $\max_{\substack{|S|=k \\ S \subset [n]}} \sum_{i \in S} p_i$  (and similarly for  $\hat{\mathbb{E}}(\hat{R}^k)$ ).

Thus, we have

$$\begin{aligned} \hat{\mathbb{E}}(\hat{R}^k) &= \sum_{i \leq k} \hat{p}_{[i]} = \max_{\substack{|S|=k \\ S \subset [n]}} \sum_{i \in S} \hat{p}_i \\ &\leq \max_{\substack{|S|=k \\ S \subset [n]}} \sum_{i \in S} p_i + \epsilon \\ &= \sum_{i \leq k} p_{[i]} + \epsilon = \mathbb{E}(R^k) + \epsilon, \end{aligned} \quad (8)$$

where at the second line (8) we have used the condition  $\|p - \hat{p}\|_1$ .

Similarly, we also have

$$\hat{\mathbb{E}}(\hat{R}^k) = \sum_{i \leq k} \hat{p}_{[i]} = \max_{\substack{|S|=k \\ S \subset [n]}} \sum_{i \in S} \hat{p}_i \quad (9)$$

$$\geq \max_{\substack{|S|=k \\ S \subset [n]}} \sum_{i \in S} p_i - \epsilon \quad (10)$$

$$= \sum_{i \leq k} p_{[i]} - \epsilon = \mathbb{E}(R^k) - \epsilon, \quad (11)$$

where at the second line (10) we have used the condition  $\|p - \hat{p}\|_1$ . The result follows.  $\square$

We now consider the recommender systems setting, where the recall is defined user-wise and averaged over the users. In this case, we have a ground truth distribution  $P \in \mathbb{R}^{m \times n}$  and its estimated

version  $\hat{P} \in \mathbb{R}^{m \times n}$ . We fix  $k$  and define the recall of user  $i$  as  $R^{k,i}$  via formula (1) where  $p \leftarrow p_{i,\cdot} \in \mathbb{R}^n$  is now the normalized version of the  $i$ th row of  $P$  (i.e.  $p_{i,j} = P_{i,j}/p_i$  with  $p_i := \sum_{j \in [n]} P_{i,j}$ ). We can similarly define the quantities relative to  $\hat{P}$ . We define the aggregated recall w.r.t. the ranking provided by  $\hat{P}$  (resp.  $P$ ) by  $\hat{R}^{k,\text{all}} := \frac{1}{m} \sum_{i \leq m} \hat{R}^{k,i}$  (resp.  $R^{k,\text{all}} := \frac{1}{m} \sum_{i \leq m} R^{k,i}$ ).

PROPOSITION A.2. Assume that  $p_i = \frac{1}{m}$  for all  $i$ , and  $\|\hat{P} - P\|_1 \leq \epsilon$ . Then we have the following excess risk bound:

$$\mathbb{E}(\hat{R}^{k,\text{all}}) \leq \mathbb{E}(R^{k,\text{all}}) + 2\epsilon. \quad (12)$$

PROOF. Let  $\epsilon_i$  be defined as  $\|p_{i,\cdot} - \hat{p}_{i,\cdot}\|_1$ . Then by Proposition A.1 we certainly have

$$\mathbb{E}(\hat{R}^{k,\text{all}}) = \mathbb{E}\left[\frac{1}{m} \sum_{i=1}^m \hat{R}^{k,i}\right] \quad (13)$$

$$= \frac{1}{m} \sum_{i=1}^m \mathbb{E}\hat{R}^{k,i} \leq \frac{1}{m} \sum_{i=1}^m \mathbb{E}R^{k,i} + 2\epsilon_i \quad (14)$$

$$\leq \mathbb{E}(R^{k,\text{all}}) + 2 \frac{1}{m} \sum_{i=1}^m \epsilon_i \quad (15)$$

$$= \mathbb{E}(R^{k,\text{all}}) + 2 \frac{1}{m} \sum_{i=1}^m \|p_{i,\cdot} - \hat{p}_{i,\cdot}\| \quad (16)$$

$$\leq \mathbb{E}(R^{k,\text{all}}) + 2 \frac{1}{m} \sum_{i=1}^m \|m p_{i,\cdot} - m \hat{p}_{i,\cdot}\| \quad (17)$$

$$= \mathbb{E}(R^{k,\text{all}}) + 2 \left\|P - \hat{P}\right\|_1 \leq \mathbb{E}(R^{k,\text{all}}) + 2\epsilon, \quad (18)$$

where at equation (14) we have used Proposition A.1 and at equation (17) we have used the assumption that  $p_i = \frac{1}{m}$  for all  $i$ . The result follows.  $\square$

## A.2 Bounding the L1 Loss

We have established above that if we can control the  $L1$  loss, then we can control the excess risk defined in terms of Recall@ $k$  for any  $k$ . To control the  $L^1$  loss, we use the following result from [2] which ultimately follows from results on Sheffé tournaments in non-parametric density estimation [1].

PROPOSITION A.3 (ADAPTATION OF PROPOSITION 2.1 IN [2]). Let  $\mathcal{H}_{m \times n, r}$  denote the set of non-negative rank  $r$  distributions over  $[m] \times [n]$ . Let  $p \in \mathcal{H}_{m \times n, r}$  be a probability distribution from which we observe  $N$  i.i.d samples. There exists an estimator  $\hat{p} \in \mathcal{H}_{m \times n, r}$  (depending on the  $N$  samples) such that for any  $\delta > 0$ , the following holds with probability  $\geq 1 - \delta$ :

$$\|p - \hat{p}\|_1 \leq 7 \sqrt{\frac{(m+n)r \log(2(m+n)rN)}{N}} + 7 \sqrt{\frac{\log(\frac{3}{\delta})}{2N}}.$$

By combining Proposition A.3 with Proposition A.2, we obtain the following:

THEOREM A.4. Let  $\mathcal{H}_{m \times n, r}$  denote the set of non-negative rank  $r$  distributions over  $[m] \times [n]$ . Let  $p \in \mathcal{H}_{m \times n, r}$  be a probability distribution from which we observe  $N$  samples. There exists an estimator  $\hat{p} \in \mathcal{H}_{m \times n, r}$  (depending on the  $N$  samples) such that for any

$\delta > 0$ , the following excess risk bound for the recall at  $k$  holds with probability  $\geq 1 - \delta$ :  $\mathbb{E}(\hat{R}^{k,\text{all}}) - \mathbb{E}(R^{k,\text{all}}) \leq$

$$14 \sqrt{\frac{(m+n)r \log(2(m+n)rN)}{N}} + 14 \sqrt{\frac{\log(\frac{3}{\delta})}{2N}}. \quad (19)$$

## References

- [1] L. Devroye and G. Lugosi. 2001. *Combinatorial Methods in Density Estimation*. Springer, New York.
- [2] Robert A Vandermeulen and Antoine Ledent. 2021. Beyond smoothness: Incorporating low-rank analysis into nonparametric density estimation. *Advances in Neural Information Processing Systems* 34 (2021), 12180–12193.